

RV64M — Multiply / Divide Extension Decoding Reference

All 14 instructions valid for RV64M. All are pure R-type. *W variants are RV64-only (32-bit operands, sign-extended 64-bit result).

opcode[6:0]		rd[11:7]		funct3[14:12]		rs1[19:15]				rs2[24:20]		funct7[31:25]=0000001		[RV64] = RV64-only (*W variants)	
Mnemonic	Fmt	opcode	funct3	funct7	Bit layout [31:0]						Operation		Notes		
RV64M — Multiply (opcode 0110011, funct7 = 0000001) — operates on full 64-bit registers															
MUL	R	0110011	000	0000001	0000001	rs2	rs1	f3	rd	0110011	rd = (rs1 x rs2)[63:0]		Lower 64 bits of product		
MULH	R	0110011	001	0000001	0000001	rs2	rs1	f3	rd	0110011	rd = (rs1 xs rs2)[127:64]		Upper 64 bits — signed x signed		
MULHU	R	0110011	011	0000001	0000001	rs2	rs1	f3	rd	0110011	rd = (rs1 xu rs2)[127:64]		Upper 64 bits — unsigned x unsigned		
MULHSU	R	0110011	010	0000001	0000001	rs2	rs1	f3	rd	0110011	rd = (rs1 xsu rs2)[127:64]		Upper 64 bits — signed x unsigned		
RV64M — Divide & Remainder (opcode 0110011, funct7 = 0000001) — 64-bit signed/unsigned															
DIV	R	0110011	100	0000001	0000001	rs2	rs1	f3	rd	0110011	rd = rs1 div(s) rs2		Signed div; rounds toward zero		
DIVU	R	0110011	101	0000001	0000001	rs2	rs1	f3	rd	0110011	rd = rs1 div(u) rs2		Unsigned division		
REM	R	0110011	110	0000001	0000001	rs2	rs1	f3	rd	0110011	rd = rs1 rem(s) rs2		Signed remainder; sign = dividend		
REMU	R	0110011	111	0000001	0000001	rs2	rs1	f3	rd	0110011	rd = rs1 rem(u) rs2		Unsigned remainder		
RV64M only — 32-bit Word variants (opcode 0111011, funct7 = 0000001) — result sign-extended to 64 bits															
MULW	R	0111011	000	0000001	0000001	rs2	rs1	f3	rd	0111011	rd = sext((rs1 x rs2)[31:0])		Lower 32-bit product, sext to 64		
DIVW	R	0111011	100	0000001	0000001	rs2	rs1	f3	rd	0111011	rd = sext(rs1[31:0] div(s) rs2[31:0])		Signed 32-bit div, sext result		
DIVUW	R	0111011	101	0000001	0000001	rs2	rs1	f3	rd	0111011	rd = sext(rs1[31:0] div(u) rs2[31:0])		Unsigned 32-bit div, sext result		
REMW	R	0111011	110	0000001	0000001	rs2	rs1	f3	rd	0111011	rd = sext(rs1[31:0] rem(s) rs2[31:0])		Signed 32-bit remainder, sext		
REMUW	R	0111011	111	0000001	0000001	rs2	rs1	f3	rd	0111011	rd = sext(rs1[31:0] rem(u) rs2[31:0])		Unsigned 32-bit remainder, sext		

Decoding key: funct7 = 0000001 on any R-type instruction signals the M extension. Base-ISA R-type uses 0000000 or 0100000 — so a single funct7 bit 25 set to 1 identifies all M instructions. Div-by-zero: DIV/DIVW returns −1 (all ones); DIVU/DIVUW returns 2^L−1 (all ones); REM/REMW returns dividend; REMU/REMUW returns dividend. Signed overflow (MIN_INT ÷ −1): DIV/DIVW returns MIN_INT, REM/REMW returns 0.